

Math 141, S10

Quiz 11

Section:

Name: *Answer*

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (5 points) Evaluate the following indefinite integrals:

(a) $\int x\sqrt{x^2+2} dx$

Let $u = x^2 + 2$, then $du = 2x dx$
i.e., $\frac{1}{2} du = x dx$

thus

$$\begin{aligned}\int x\sqrt{x^2+2} dx &= \int \sqrt{u} \cdot \frac{1}{2} du \\ &= \frac{1}{2} \cdot \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + C \\ &= \frac{1}{3} u^{\frac{3}{2}} + C \\ &= \frac{1}{3} (x^2+2)^{\frac{3}{2}} + C\end{aligned}$$

(b) $\int \cos \theta \sin^4 \theta d\theta$

Let $u = \sin \theta$, then $du = \cos \theta d\theta$
thus

$$\begin{aligned}\int \cos \theta \sin^4 \theta d\theta &= \int u^4 du \\ &= \frac{u^5}{5} + C \\ &= \frac{(\sin \theta)^5}{5} + C\end{aligned}$$

Problem 2. (5 points) Evaluate the following definite integrals:

(a) $\int_1^{e^2} \frac{\ln x}{x} dx$

Let $u = \ln x$, then $du = \frac{1}{x} dx$

When $x=1$, $u(1) = \ln 1 = 0$

$x=e^2$, $u(e^2) = \ln e^2 = 2 \ln e = 2$

Thus, $\int_1^{e^2} \frac{\ln x}{x} dx = \int_0^2 u du$

$$\begin{aligned}&= \left[\frac{u^2}{2} \right]_0^2 \\ &= \frac{2^2}{2} - 0 \\ &= 2\end{aligned}$$

(b) $\int_0^1 x e^{-x^2} dx$

Let $u = -x^2$, then $du = -2x dx$
i.e., $-\frac{1}{2} du = x dx$

When $x=0$, $u(0) = 0$

$x=1$, $u(1) = -1$

Thus,

$$\begin{aligned}\int_0^1 x e^{-x^2} dx &= \int_0^{-1} e^u \cdot \left(-\frac{1}{2}\right) du \\ &= -\frac{1}{2} [e^u]_0^{-1} \\ &= -\frac{1}{2} (e^{-1} - e^0) \\ &= -\frac{1}{2} (e^{-1} - 1) \\ &= \frac{1}{2} \left(1 - \frac{1}{e}\right)\end{aligned}$$